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The Role of Solvent Polarity in the Free Energy of Transfer of Amino Acid Side Chains from Water to Organic Solvents*

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The transfer free energies of amino acid side chains from water to *N*-methylacetamide have been determined and compared with those obtained from other model systems. Although the process of transfer from water to *N*-methylacetamide represents transfer from a lower dielectric phase to a higher dielectric phase, the transfer free energies of most of the amino acid side chains are nearly the same as those obtained from the water to ethanol system. Among the apolar side chains studied, only the transfer free energies of methionine and the aromatic side chains are apparently influenced to some extent by the polarity of the organic solvent phase. The transfer free energies of the neutral polar side chains also exhibit significant dependence on solvent polarity. The van't Hoff plots for most of the apolar side chains exhibit nonlinear curves, indicating that the enthalpy of transfer from water to *N*-methylacetamide is temperature-dependent. It is suggested that to assess the contribution of the hydrophobic free energy to the stability of globular proteins, it is probably not necessary to account for variation in the internal environment of the protein.

Ever since Kauzmann's prediction (1) that hydrophobic interactions between nonpolar side chains play an important role in the structural stability of globular proteins, there have been many attempts to quantify the hydrophobicities of amino acids (2–8). These approaches are based on thermodynamic treatment of the distribution of amino acids between water and organic solvents. The main criterion in selecting the organic phase was its polarity or dielectric constant (5, 8).

Assuming that the interior of a globular protein is analogous to an ethanol environment, Tanford and co-workers (2, 3) estimated the transfer free energies of side chains from ethanol to water from solubility measurements. Using the same rationale, but assuming that the interior of a protein is analogous to a hexane environment, Fendler *et al.* (6) determined the free energies of transfer from hexane to water. In another approach based on the surface tension of amino acid solutions, Bull and Breese (4) calculated the hydrophobicities of amino acids and showed that their values were comparable to those of Nozaki and Tanford (3).

Although these approaches provide an estimate of the hydrophobicities of various amino acid side chains, there is considerable skepticism as to their usefulness in ascertaining the contribution of hydrophobic free energy to the stability of proteins (9). The lack of confidence in the hydrophobicity values obtained from these methods arises because of uncertainties involved in assessing the internal environment of a protein (7). Since the interior of a protein is neither a dielectric continuum (10) nor homogeneous in composition, it is being argued that no single solvent system can adequately represent the microenvironment experienced by nonpolar side chains in the interior of a protein (7). Recognizing the fact that regions of different polarities would be present, Nandi (5) studied the transfer free energies of certain side chains from several solvents of varying polarities to water. Interestingly, no significant difference in free energies of transfer were observed.

Current evidence suggests that the transfer free energies of side chains are independent of the nature of the organic phase. However, it must be noted that the phases studied have been less polar than water; further, the dielectric constants of the solvents used have not differed very much. It is possible that small differences in polarity may not significantly influence the transfer free energies. To conclusively demonstrate whether or not solvent polarity influences the hydrophobic free energies of apolar side chains, one should study the transfer free energies from water to solvents which differ widely in polarity. We have studied the thermodynamics of transfer from water to *N*-methylacetamide (dielectric constant 191) and compared our results with those from other systems, *e.g.* water to ethanol (dielectric constant 25). Here we report that the transfer free energies from water to *N*-methylacetamide are not appreciably different from those of the water to ethanol system.

MATERIALS AND METHODS

L-Amino acids were purchased from Sigma and used without further purification. *N*-Methylacetamide (>99%) was purchased from Aldrich. All other reagents were of reagent grade. Distilled and deionized water was used in all experiments.

The approach used for solubility measurements was similar to that described elsewhere (3). Known amounts of solvent and amino acid were placed in glass vials with screw caps. Care was taken that the amount of amino acid in each vial was more than the maximum solubility under the conditions employed. A small Teflon-coated magnetic bar stirred each vial. In the case of tryptophan and cysteine, to prevent air oxidation during the equilibration, the contents of the vials were deareated by suction and flushed with nitrogen. Vials were kept at least for 48 h in a water bath held to within ±0.05 °C. Afterwards, the contents were allowed to settle for 30 min. An aliquot of the supernatant was removed using a syringe fitted with a filtration unit which contained two discs of 0.2-μ membrane filter (Gelman). The samples were diluted as required with water and the amino acid concentrations were determined by the 2,4,6-trinitrobenzenesulfonic acid (TNBS[1]) method (11). Instead of using a standard curve for

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[1] The abbreviations used are: TNBS, 2,4,6-trinitrobenzenesulfonic acid; NMA, *N*-methylacetamide.

leucine, we determined the extinction coefficients of the TNBS derivatives at 420 nm and used these to determine the solubilities. For each amino acid, measurements were made at least in duplicate; the variation with duplicates was negligible. The free energies of transfer from water to N-methylacetamide were calculated using the relationship

$$\Delta G_{tr} = (-RT \ln S^{AA}_{NMA}/S^{AA}_{H_2O}) - (-RT \ln S^{Gly}_{NMA}/S^{Gly}_{H_2O})$$

where S^{AA}_{NMA} and $S^{AA}_{H_2O}$ are solubilities of amino acids in NMA and water, respectively; S^{Gly}_{NMA} and $S^{Gly}_{H_2O}$ are solubilities of glycine in NMA and water, respectively.

RESULTS

The results at 37 °C are shown in Table I. Since the melting point of NMA is 28 °C, solubility measurements were possible only above 30 °C. In preliminary experiments, we have studied the solubilities of amino acids in aqueous NMA solutions containing 0 to 80% NMA (w/w). Plots of solubility against NMA concentration were curved for all the amino acids studied (data not illustrated). Extrapolation either by linear regression or nonlinear curve-fitting to obtain the solubility in 100% NMA resulted in ambiguous values. In subsequent studies, we measured the solubilities in water and 100% NMA directly. Since the solubilities in NMA ranged from 3 to 70 mM, accurate determination of concentrations by the TNBS method (11, 12) was possible. Further, since the TNBS reaction is specific for primary amino groups, no interference from NMA was observed in the TNBS procedure.

As shown in Table I, the transfer free energies, ΔG_{tr}, are all negative, indicating that the transfer process is thermodynamically favorable both for polar and nonpolar side chains. For comparison, the ΔG_{tr} values for water to ethanol (13), water to surface (4), and water to hexane (6) are also shown in Table I. Although the transfer free energies of certain side chains for water to NMA are slighly higher than those for water to ethanol, the differences are not very significant. The two sets of data exhibit a correlation coefficient of $r = 0.82$. The ΔG_{tr} values of four side chains, namely, isoleucine, asparagine, aspartate, and glutamate show the greatest differences. When the ΔG_{tr} values for these side chains are ignored, the correlation coefficient becomes 0.944. In the water to ethanol system, the ΔG_{tr} of the isoleucine side chain is higher (more negative) than that of leucine, whereas in the present study the values exhibit the opposite trend (Table I). However, ΔG_{tr} values for these two side chains obtained from the water to surface and water to hexane systems (Table I) are in accordance with our results. Since the solubility of isoleucine in water is more than that of leucine, a lower hydrophobicity value would be expected for isoleucine compared to leucine. Hence, it is probable that the hydrophobicity value for isoleucine obtained from the water to ethanol system (3) may be erroneous and may need re-evaluation.

TABLE I

Free energies of transfer of amino acid side chains from water to organic solvents at 37 °C

Amino acid	$-\Delta G_{tr}$ ($H_2O \rightarrow$ NMA)	$-\Delta G_{tr}$ ($H_2O \rightarrow$ ethanol)[a]	$-\Delta G_{tr}$ ($H_2O \rightarrow$ surface)[b]	$-\Delta G_{tr}$ ($H_2O \rightarrow$ hexane)[c]
	kcal/mol			
Glycine	0	0	0	0
Alanine	0.666	0.801	0.208	−0.436
Leucine	2.550	2.150	2.560	2.800
Isoleucine	2.294	3.170	2.350	2.756
Valine	1.685	1.841	1.623	1.750
Methionine	2.324	1.633	1.529	
Phenylalanine	3.016	2.881	2.423	2.590
Tyrosine	3.576	2.673	2.330	
Tryptophan	4.270	3.817	2.090	
Cysteine	1.285	1.477	0.468	
Serine	0.533	−0.031	0.406	−0.800
Threonine	0.774	−0.031	0.540	−0.041
Histidine	0.658	0.801	0.125	0.600
Arginine	0.624	0.780	0.125	−1.500
Lysine	1.236	1.602	0.364	0.260
Aspartic acid	2.389	0.582	0.208	1.290
Glutamic acid	1.896	0.593	0.312	1.020
Aspargine	1.358	−0.010	−0.083	
Glutamine	0.205	−0.100	−0.166	

[a] Taken from Ref. 15.
[b] Taken from Ref. 4.
[c] Taken from Ref. 6.

The effect of temperature on the transfer free energies of apolar side chains from water to NMA is given in Table II. A van't Hoff plot for the transfer of these side chains from water to NMA is shown in Fig. 1. The van't Hoff plots for most of the side chains are nonlinear in the temperature range studied. The curves for alanine, valine, isoleucine, and phenylalanine exhibit a maximum at around 37 to 40 °C, where ΔH_{tr} is zero. This suggests that for these side chains the hydrophobic transfer free energy from water to NMA is maximum at around 37 to 40 °C; above and below this temperature, the enthalpy changes are negative and positive, respectively. In the case of tryptophan, tyrosine, methionine, and leucine, no distinct maxima were observed. However, the maximum for these side chains may occur at temperatures lower than 30 °C.

It has been reported that the enthalpy change for the transfer of apolar side chains from water to organic solvents (5) or from water to surface (4) is independent of temperature in the range 0 to 55 °C. However, our results show that for the transfer from water to NMA, the ΔH_{tr} of most of the apolar side chains is dependent on temperature in the same range (Fig. 1). Further, Bull and Breese (4) have reported that the enthalpy change for the transfer of the leucine side chain from water to surface is about +0.56 kcal/mol in the temperature range 30 to 55 °C. Similarly, Nandi (5) obtained a ΔH_{tr} value of about +0.5 kcal/mol for the transfer of the leucine side chain from water to 1-hexanol or 1-octanol in the temperature range 0 to 40 °C. However, from linear regression analysis of the data in Fig. 1 for leucine, we obtained a ΔH_{tr} value of −0.27 kcal/mol in the 30 to 44 °C temperature range. The negative ΔH_{tr} obtained in our studies is in disagreement with the positive ΔH_{tr} values reported earlier (4, 5). However, the ΔG_{tr} values for leucine obtained in this study and in the previous studies (4, 5) are almost the same (Table I). This suggests that while the ΔG_{tr} values for the transfer from water to organic solvent are apparently independent of the nature of the solvent, the ΔH_{tr} values are not.

DISCUSSION

The rationale behind selecting N-methylacetamide as the organic phase is as follows. The dielectric constant of N-

TABLE II

Free energies of transfer of apolar amino acid side chains from water to N-methylacetamide at various temperatures

Amino acid	$-\Delta G_{tr}$ 30 °C	37 °C	44 °C
		kcal/mol	
Alanine	0.420	0.666	0.574
Leucine	2.476	2.549	2.540
Isoleucine	2.199	2.441	2.368
Valine	1.571	1.840	1.796
Methionine	2.258	2.324	2.251
Phenylalanine	2.521	3.016	3.037
Tyrosine	3.515	3.576	3.531
Tryptophan	4.221	4.270	4.210

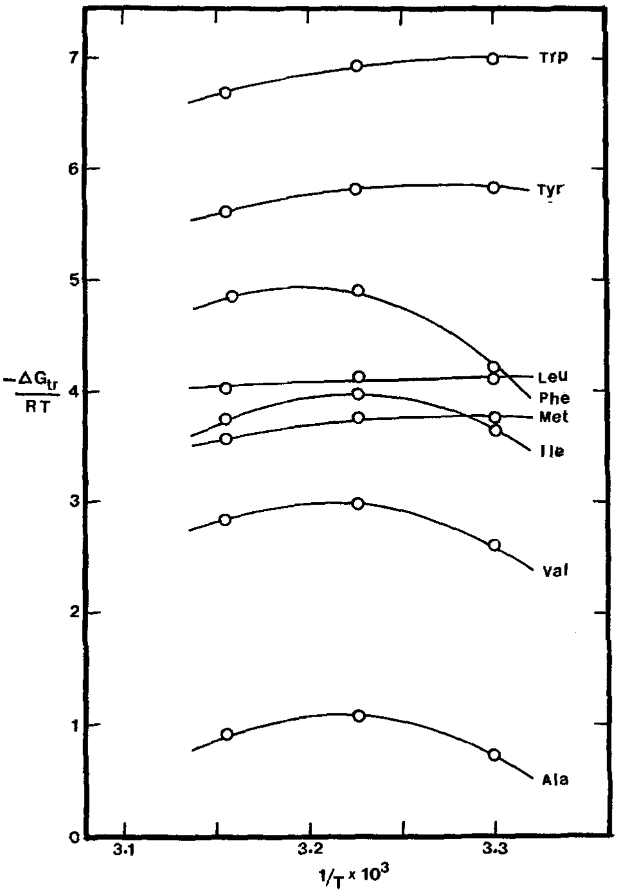

FIG. 1. **van't Hoff plot for the transfer of apolar amino acid side chains from water to *N*-methylacetamide.**

methylacetamide at 32 °C is 191 and the dipole moment is about 4.39 (14); these values are higher than those of other organic solvents. The highly hydrogen-bonded structure of *N*-methylacetamide may mimic the densely packed cohesive interior of a protein. Further, *N*-methylacetamide may adequately represent the backbone peptide units buried in the interior which may be involved in dipole-induced dipole interaction with nonpolar side chains; such interactions have been postulated to exist in the interior of proteins (15, 16).

The data in Table I mainly reflect the influence of solvent polarity on the transfer free energies of side chains from water to various nonaqueous phases. While the water to surface (4), water to ethanol (3), and water to hexane (6) systems represent the transfer of amino acid side chains from a higher dielectric phase to a lower dielectric phase, the water to NMA system represents the transfer process from a lower dielectric phase to a higher dielectric phase. Although these sytems represent radically opposite conditions, the transfer free energies of alanine, leucine, valine, phenylalanine, tryptophan, cysteine, histidine, arginine, and lysine side chains from water to NMA and water to ethanol are very similar. Similarly, the transfer free energies of leucine, isoleucine, valine, serine, and threonine side chains from water to NMA are very similar to those from water to surface (4). Subtraction of the transfer free energy of valine from that of leucine indicates that the ΔG_{tr} of a methylene group from water to NMA is about −865 cal/mol. This is in agreement with the reported value of about −800 cal/mol for transfer from water to other organic phases (5, 17).

Although the overall differences in the transfer free energies of side chains between the three sets of data (Table I) are not significant, the ΔG_{tr} of certain side chains such as methionine, phenylalanine, tyrosine, tryptophan, aspartate, glutamate, asparginine, and glutamine apparently increase with the polarity of the organic phase. The greatest effect is seen for aspartate, asparagine, glutamate, and glutamine. In spite of their strong interaction with water, these polar side chains tend to prefer the *N*-methylacetamide phase. The probable reason may be the strong hydrogen bonding potential of NMA. Because of the potential for hydrogen bond formation between NMA and these polar side chains, transfer from water to NMA may not result in any net loss of hydrogen bonding free energy. However, this may not be the case with ethanol, which possesses lower hydrogen bonding potential than NMA.

Among the apolar side chains, only the ΔG_{tr} of methionine and the aromatic side chains are influenced by solvent polarity (Table I). Since these apolar side chains are more polarizable than other apolar side chains, the observed effect may be attributed to dipole-induced dipole interaction between NMA and these side chains. The existence of such interactions in the interior of proteins has been postulated (15, 16). Each of such polar-apolar interactions may provide an additional −0.5 kcal/mol for the stability of globular proteins (15). It should be noted that ethanol contains only one polar group, whereas NMA contains two polar groups. The difference in the ΔG_{tr} values of aromatic side chains between water to NMA and water to ethanol systems may be due to one additional polar-apolar interaction.

The results presented here demonstrate that the free energy of transfer of amino acid side chains from water to a nonaqueous phase is not very much dependent on the polarity of the chosen reference phase. It seems to be solely the manifestation of the unique three-dimensional structure of water. Based on our results, we suggest that in order to assess the magnitude of the contribution of hydrophobic free energy for the stability of globular proteins, it is not necessary to account for variation in the internal environment of the protein. Since the variations in the dielectric constant of various microregions in the interior of the protein may range from 40 to 80 (18), these differences will not significantly affect the hydrophobic free energies of apolar side chains.

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